PS Vita Anki-Compatible Handheld

Finalized Product Direction

Core Vision

Build a native PS Vita homebrew application that functions as:

> an Anki-compatible handheld review console with integrated gamification

The project focuses on:

- reliable reviewing,
- scheduling fidelity,
- tactile controls,
- portable study sessions,
- psychologically engaging progression systems.

Desktop Anki remains the canonical ecosystem.

The Vita becomes:

> the optimized review frontend.

Final Product Identity

The application is NOT:

- a full desktop Anki replacement,
- a Qt/webview port,
- a browser-based clone,
- an addon runtime environment.

The application IS:

- an Anki-compatible review client,
- a handheld-focused SRS console,
- a gamified productivity device.

Compatibility Goals

Full Review Compatibility

The project will aim for compatibility with standard Anki review workflows:

- standard note types
- cloze deletion
- images
- audio
- scheduling continuity
- deck organization
- review history synchronization

Desktop Anki collections should remain usable without requiring users to redesign their study workflow.

Explicit Non-Goals

The project will NOT attempt:

- arbitrary addon execution
- arbitrary JavaScript template execution
- full HTML/CSS browser parity
- desktop dashboard replication
- full Qt/webview rendering

Reason:
these dramatically expand complexity and would likely make the project unsustainable on Vita hardware.

System Architecture

Desktop Side

A Python desktop companion application acts as:

the compatibility and synchronization bridge

Responsibilities:

- parse Anki collections
- preprocess templates
- package decks/media
- export Vita-compatible review data
- import review logs back into Anki
- preserve scheduling fidelity

The desktop layer absorbs upstream Anki complexity.

Vita Side

The Vita client is intentionally lightweight and optimized for responsiveness.

Responsibilities:

- render cards
- accept grades
- display media
- run gamification systems
- persist local review state
- export review logs

The Vita focuses on:

- review speed,
- ergonomics,
- engagement.

Rendering Philosophy

Supported:

- text
- cloze deletions
- images
- audio
- basic HTML subset

Possibly later:

- limited CSS support

Unsupported:

- arbitrary JS
- browser-like template execution
- desktop addon rendering logic

The rendering system should prioritize:

- stability,
- speed,
- predictability.

Scheduling Architecture

Scheduling fidelity is considered mission-critical.

Preferred Model

Desktop Anki remains scheduling authority.

Workflow:

1. Vita records:
    - grades
    - timestamps
    - response timing
2. Desktop importer:
    - applies official scheduling logic
    - updates collection

Advantages:

- prevents scheduler drift
- preserves compatibility
- avoids immediate FSRS reimplementation

Possible later upgrade:

- closer local FSRS parity

Synchronization Model

Initial synchronization avoids direct AnkiWeb integration.

Workflow:

Desktop Anki
↓
Exporter
↓
Vita package
↓
Review session
↓
Review logs
↓
Desktop importer

Transfer methods:

- USB
- SD card
- optional local Wi-Fi later

Direct AnkiWeb support is deferred because it introduces:

- authentication complexity
- media reconciliation
- protocol maintenance
- conflict resolution

Input Design

The Vita’s built-in hardware is sufficient.

Primary controls:

- D-pad
- face buttons
- shoulder buttons
- touchscreen

Possible mapping:

X = Good
O = Again
□ = Hard
△ = Easy

R = Flip card
L = Replay audio

Goals:

- low-friction reviewing
- reduced thumb fatigue
- one-handed usability options
- instant interaction responsiveness

No external controller support is required for MVP.

Gamification System

Gamification is now a first-class feature, not an afterthought.

The goal is:

> increase consistency and session engagement without undermining actual learning quality

The system should reward:

- consistency,
- review completion,
- healthy study habits,
- sustained retention behavior.

Planned Gamification Features

Streak System

Track:

- daily reviews

- consecutive study days
- milestone streaks

Features:

- streak freeze tokens
- streak recovery mechanics
- milestone celebrations

Experience / Leveling

Users gain XP for:

- reviews completed
- difficult cards mastered
- daily goals achieved
- streak maintenance

Levels unlock:

- cosmetic themes
- UI customization
- achievement badges

Gamification should remain mostly cosmetic to avoid distorting scheduling behavior.

Daily Missions

Examples:

- review 50 cards
- clear all due reviews
- complete study session without interruptions
- finish difficult deck

Purpose:

- encourage session completion
- reduce procrastination

Achievement System

Possible achievements:

- 1,000 reviews
- 30-day streak
- mastering difficult deck
- perfect review day
- late-night study session
- review marathon

Achievements should reinforce habit formation.

Session Statistics

Display:

- review accuracy
- cards/hour
- retention estimates
- daily progress
- streak progress
- session duration

The Vita format is well-suited for compact motivating dashboards.

Unlockable Themes

Themes become progression rewards.

Examples:

- retro themes
- minimalist themes
- Onigiri-inspired themes
- OLED-focused dark themes
- animated backgrounds (lightweight only)

This preserves motivation without interfering with learning integrity.

Onigiri Compatibility Position

The project supports:

- standard Anki collections
- selected Onigiri-inspired themes
- gamification concepts inspired by Onigiri

The project does NOT support:

- executing Onigiri addon runtime logic
- desktop-only dashboard systems
- arbitrary addon JS execution

Instead:

- themes and aesthetics are reimplemented natively for Vita.

Technical Stack

Desktop

Language:

- Python

Likely tools:

- Anki Python API
- sqlite3
- media packaging tools

Vita

Language:

- C++
- vitaSDK

Likely systems:

- SQLite
- lightweight renderer
- texture cache
- audio/image loaders
- asynchronous media loading

MVP Scope (Version 0.1)

Included

- deck import/export
- reviewing
- cloze support
- image/audio support
- local persistence
- review logging
- streak system
- XP system
- achievements
- session stats
- unlockable themes
- suspend/resume support

Excluded

- note editing
- deck editing
- addon execution
- arbitrary JS
- full HTML/CSS parity
- direct AnkiWeb sync
- complex social systems

Strategic Positioning

The project occupies a real niche:

> a dedicated handheld spaced repetition console

The Vita hardware naturally supports:

- tactile reviewing,
- portable studying,
- low-distraction sessions,

- fast suspend/resume behavior,
- long battery life.

The project succeeds if it becomes:

> the best physical-button Anki review experience

The project fails if it attempts:

> full desktop feature parity on unsupported hardware